# EEG Signal Classification in Neurological Disorders

Academic Year 2023–2027

## 1. Project Information

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| **Project Title** | EEG Signal Classification in Neurological Disorders |
| **Domain** | Deep Learning / Biomedical Signal Processing / Blockchain |
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| **Presentation Year** | 2026 |

## 2. Abstract

This project presents an intelligent, automated system for the classification of Electroencephalogram (EEG) signals to detect and diagnose neurological and mental health disorders. EEG signals, which capture brain electrical activity, are inherently complex and traditionally require expert neurologists for manual interpretation. The proposed system addresses these limitations by employing a Long Short-Term Memory (LSTM)-based deep learning model for accurate multi-class classification of disorders including Anxiety, Obsessive-Compulsive Disorder (OCD), Addictive Disorders, and Healthy conditions. To enhance transparency and clinical trust, a Large Language Model (LLM)-based explanation module is integrated to generate human-readable interpretations of predictions. Additionally, a private Ganache blockchain is used to ensure tamper-proof and secure storage of classification results.

## 3. Problem Statement

The manual analysis of EEG data faces several critical challenges:

- EEG data is highly complex and difficult to interpret manually, requiring specialized expertise.
- Manual diagnosis depends heavily on expert neurologists, making it inaccessible in resource-limited settings.
- Large volumes of EEG recordings significantly increase the time and effort required for analysis.
- Human error during manual interpretation can lead to misdiagnosis and incorrect treatment.
- Multiple neurological disorders exhibit overlapping EEG patterns, making differentiation difficult.

- Existing automated systems often lack explainability, transparency, and secure data storage.

## 4. Proposed Solution

To address the above challenges, the following approach is proposed:

- Develop an automated EEG classification pipeline using a Deep Learning-based LSTM model trained for multi-class disorder detection.
- Preprocess raw EEG signals through filtering, normalization, and segmentation to improve signal quality and model accuracy.
- Integrate an LLM-based explanation module to generate clear, human-readable clinical descriptions of the model's predictions.
- Implement a private Ganache blockchain to store classification results securely and ensure tamper-proof record keeping.
- Design a web-based client–server application using Flask for real-time EEG upload, prediction, and result visualization.

## 5. System Architecture

The system is organized into four functional layers:

### 5.1 User Interface Layer

A web-based interface built using HTML and CSS allows users to upload EEG data files, view classification predictions, and access LLM-generated explanations in real time.

### 5.2 AI & Processing Layer

EEG signals are preprocessed and passed into the LSTM deep learning model for multi-class classification. An integrated LLM module then generates prediction explanations to support clinical decision-making.

### 5.3 Backend Application Layer

Developed using the Flask framework, this layer manages file uploads, preprocessing pipelines, LSTM model execution, LLM explanation generation, and communication with the blockchain layer.

### 5.4 Blockchain & Storage Layer

Classification results are securely stored using a private Ganache blockchain, ensuring data integrity, transparency, and tamper-proof record keeping for each diagnostic session.

## 6. Technologies Used

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| **DEEP LEARNING MODEL** | LSTM (Long Short-Term Memory) |
| **EXPLANATION MODULE** | LLM (Large Language Model) |
| **BACKEND FRAMEWORK** | Flask (Python) |
| **FRONTEND** | HTML, CSS |
| **BLOCKCHAIN** | Ganache (Private Ethereum) |
| **SIGNAL PROCESSING** | Filtering, Normalization, Segmentation |

## 7. Conclusion

The proposed EEG signal classification system successfully automates the detection of multiple neurological and mental health disorders using an LSTM-based deep learning model. The system significantly reduces manual diagnostic effort, improves classification accuracy, and handles the complex temporal nature of EEG time-series data effectively. By integrating an LLM-based explanation module, the system enhances clinical transparency and trustworthiness. The use of a private blockchain ensures secure, immutable storage of all classification outcomes. Overall, this project delivers a reliable and intelligent framework that combines deep learning, explainable AI, and blockchain technology for automated neurological disorder detection.

## 8. Future Scope

- Extend the system to support real-time EEG signal monitoring and live disorder detection.
- Expand the classification model to detect additional neurological and psychiatric conditions with improved accuracy.
- Integrate with hospital management systems and cloud-based platforms for better scalability and accessibility.
- Apply advanced AI techniques for predictive analytics and early-stage risk assessment.
- Develop mobile application support and integrate IoT-based EEG devices for remote and clinical healthcare environments.